Local Conference of Youth Hong Kong 2024

Beyond 1.5°C: A Climate-Resilient Future for All

Youth Collective Position Papers

LCOY | a project of YOUNGO OFFICIAL YOUTH CONSTITUENCY OF THE UNFCCC

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Introduction

In September 2023, Hong Kong experienced a "once-in-500-years" rainstorm that left one person dead, 117 injured, and extensive damage to vehicles, shopping centers, and properties due to flooding. This extreme weather event prompted the Hong Kong government to issue its first and longest-ever extreme weather warning, leading to school closures.

The stark reality presented in the World Meteorological Organization's 2023 report is deeply concerning - it confirms that 2023 is on track to be the hottest year on record, with temperatures 1.4°C above the pre-industrial baseline. This sobering finding underscores the urgency required to address the worsening climate crisis, even as global leaders strive to limit temperature rise to the crucial 1.5°C threshold.

Co-organised by CarbonCare InnoLab (CCIL), Generation Climate Hong Kong (GenC), Hong Kong Youth for Climate Actions (HKYCA) and the Network of Environmental Student Society (NESS), LCOY HK 2024 is designed to address the pressing challenges of the climate emergency, promoting climate justice and inspiring the development of a climate-resilient future for Hong Kong.

LCOY HK 2024 provides a dynamic platform for youth to engage with climate change issues, connect with peers, and empower each other to become climate leaders, creative problem solvers, advocates, and networkers. Through a series of seminars, panel discussions, tours, and interactive workshops under the theme of "Beyond 1.5°C: A Climate-Resilient Future for All", the conference focuses on encouraging collaborative solutions and innovative actions. It also hopes to strengthen networks among young people as well as inspire the determination to continue climate actions, in preparation for COP 29 at the end of the year.

In this position paper, participants at LCOY HK have articulately expressed their perspectives and recommendations, advocating for a climate-resilient Hong Kong. The paper delves into crucial topics including **(1) Action for Climate Empowerment, (2) Climate Justice and Human Right, (3) Climate Finance, (4) Phase out of Fossil Fuels and Renewable Energy Development (5) Climate Resilience (6) Technology and Innovation(7) Nature, Biodiversity and Oceans (8) Politics, Policy Making and Governance and (9) Circular Economy.** These valuable insights will contribute to the broader youth statement coordinated by YOUNGO under the UNFCCC, influencing the global youth climate discourse at COY 19 and COP 29 in 2024.

Summary of Recommendations

Action for Climate Empowerment

1. Prioritise environmental and sustainability education in educational policy
2. Strengthens climate education at schools
3. Enhance local public participation and improve accessibility of relevant information

Climate Justice and Human Right

1. Reducing residents' exposure by increasing shaded areas on streets and strengthening cooling center services.
2. Bolstering public health policies to better address the mortality risks associated with high temperatures.
3. Develop resilient health systems and infrastructure, including for water/sanitation.
4. Promote universal health coverage and social protection, attentive to vulnerable groups, with accessible early-warning information.
5. Target climate action opportunities that achieve co-benefits, reducing ill-health, building community resilience, and alleviating poverty.
6. Extend and expand the "Assistance Programme to Improve the Living Environment of Low-Income Subdivided Unit Households".
7. Implement a registration system for subdivided flats.
8. Prioritise the residents living in subdivided flats in the public housing allocation system.
9. Consider climate resilience in future interim housing, public housing, and light public housing projects.
10. Extreme weather information and relevant health information available in different languages like Bahasa Indonesia, Hindi, Nepali, Punjabi, Tagalog, Thai, Urdu, and Vietnamese.
11. Information about temporary shelters for extreme weather available in different languages.
12. At least one staff member stationed at a temporary shelter for extreme weather can offer necessary help for ethnic minorities in their language.

Climate Finance

1. Implement a carbon tax in Hong Kong for emissions control, with exemptions for SMEs and high-emission sectors.
2. Boost Hong Kong's Green Bond funding for climate adaptation to enhance urban resilience in light of recent extreme weather events.

Phase out of Fossil Fuels and Renewable Energy Development

1. Setting ambitious renewable energy targets and a robust roadmap.
2. Engagement with the private sector to accelerate the deployment of renewable energy.
3. Maximise the use of existing resources and infrastructure.

Climate Resilience

1. Ensure outdoor workers have rest breaks and protective gear for safety.
2. Simplify Labor Tribunal claims process for quick resolutions.
3. Extend delivery times, provide heatwave insurance for couriers.
4. Offer tailored welfare support for low-wage food delivery workers.
5. Mandate compensation and insurance for self-employed couriers.
6. Provide anti-heat uniforms, improve equipment, and use data for cooler routes.

Technology and Innovation

1. Develop a robust talent pipeline.
2. Increase awareness and training for researchers to commercialise innovative research.
3. Utilise relevant technologies to increase climate resilience.

Nature, Biodiversity and Oceans

1. Align with International initiatives and policies, Hong Kong must expand marine and coastal protected areas to cover 30% of its waters.
2. Regular monitoring and engagement of local communities are essential for effective land conservation strategies.
3. Integrating Nature-based Solutions (NbS) into climate strategies enhances coastal defense, mitigates climate change, and boosts biodiversity and ecosystem resilience.
4. Use of natural areas to lower urban temperatures, enhancing the comfort and livability of cities.
5. Publicise environmental education by integrating comprehensive climate and nature literacy into the school curriculum.

Politics, Policy Making and Governance

1. Implement comprehensive consultation methods innovatively to increase public engagement in sustainability and climate change policy-making.
2. Develop an interdepartmental coordination mechanism within the Hong Kong Government to address the 'diffusion of responsibility' phenomenon.

Circular Economy

1. Refine the Producer Responsibility Schemes (PRS) by promoting Design for Environment (DfE), introducing mandatory Fee Modulation, and transitioning to Advanced Fee Modulation for precise incentives.
2. Implement MSW charging to drive waste reduction, focusing on major sources like food waste, plastic, and paper, aligning efforts with PRS enhancements.

Action for Climate Empowerment

Introduction

The rapid development and industrial revolution have resulted in a catastrophic rising global temperature that started in the 1880s and has been accelerated since 1975 (NASA Earth Observatory, n.d.). Realizing the far-reaching effects brought by global climate change, the Intergovernmental Panel on Climate Change (IPCC) was established in 1988 (IPCC, n.d.) and laid a foundation for future international cooperation. We firmly believe that empowering the next generation through education and training is the most important and effective way to mitigate the issue since education helps people understand and address the impacts of the climate crisis, empowering them to act as agents of change (UNESCO, n.d.). In addition, the solution must be in line with the United Nations Sustainable Development Goals to promote long-term sustainability.

Recently in Hong Kong, more non-governmental organizations such as student-led and youth-led societies have been established to call for public action to combat climate change. The Environmental Protection Department has introduced numerous educational facilities to increase public awareness like T-PARK, O-PARK, and WEEE-PARK (GovHK, 2024). Plus, the Environment and Conservation Fund was set up to provide funding to different organisations or research institutes to promote local environmental protection activities (Environment and Conservation Fund, n.d.). The government also encourages strengthening the learning materials and enhancing inter-university cooperation on climate change education (Environment Bureau, 2021).

Nonetheless, the effect of policies has not been maximised due to several problems and challenges faced by the local society.

1. Lack of Comprehensive Education Syllabus and Guidance
   Though climate education is imposed in compulsory education, the coverage is not comprehensive as most of the content is focused on the physical facts about climate change such as the cause of climate change, and efforts done by global society, but not how local youth can participate and voice. Additionally, climate education is only limited to certain students i.e. Senior Geography students, leading to fewer students realising the need to cope with climate change. Because of the limited syllabus, not all educators are trained, and not much guidance is provided to clarify the needs and curriculum of climate education. For those who have not received any training, they cannot conduct climate education. As a consequence, the lack of support is hindering the whole climate education development.

2. Lack of Public Awareness and Participation
   Most of the citizens are not willing to participate in mitigating climate change as they do not realise the urgency of the issue, reflecting the low public awareness and participation situation. Besides, the scale of the current environmental protection campaign is not large enough to let all citizens take part as most climate actions only involve officials or people who are interested in solving the problem instead of the public.

Recommendations

1. Prioritise environmental and sustainability education in educational policy
   Harnessing education is paramount in propelling movements towards a sustainable future. As declared by the UNESCO, environmental education must be a core curriculum component by 2025 (UNESCO, 2021). Several countries/regions, such as Italy, Korea, and Taiwan have already incorporated environmental education as an essential part of their formal school curricula across various educational levels. Hong Kong significantly lags behind as environmental education is not prioritised, being merely considered a minor component within our educational system's values education framework (Curriculum Development Council, 2021). We urge the Hong Kong Government to take the following actions to review the status of environmental education in Hong Kong:

   I. Set up a new institution in the Education Bureau (EDB) designated for overseeing and coordinating the implementation of environmental education.

   II. Update the Guidelines on Environmental Education in Schools which has remained stagnant since 1999.

   III. Adopt a goal-oriented approach. Formulate Hong Kong's first environmental education/education for sustainable development action plan (Ma, 2019). Clearly specify the detailed environmental education aims in subject curricula and the criteria of success, especially for compulsory subjects such as primary General Studies, junior secondary Science, and senior secondary Citizenship and Social Development.

   IV. Review of school curriculum, shifting the emphasis of environmental education from climate facts to outdoor education and climate actions to counteract citizens' passive attitude towards pro-environmental behaviour changes (劉彥汶, 2021).

   V. Develop training programs for educators to raise their awareness on climate change, while equipping subject teachers with the knowledge and skills necessary to effectively teach environmental concepts in the formal curricula.

2. Strengthen climate education at schools
   As school education is an essential part of climate empowerment, we recommend that local schools:

   I. Encourage student-led actions
   Schools should support the foundation of student-led societies like Climate Change Society and Environmental Conservation Society by providing related resources and guidance. It allows students to organise projects or campaigns, empowering other students with the knowledge and responsibilities to combat climate change.

   II. Promote interactive learning
   It is suggested that schools integrate climate change topics into hands-on activities or field trips. For example, through participating in field trips to mangrove, students can observe the environment and biodiversity, gaining knowledge about the environment in the real world instead of learning from words or photos. Meanwhile, follow-up discussion allows students to build up senses of empathy and responsibilities

III. Update school-based curriculum based on the guidance regularly
To equip students with the knowledge and skills required to address climate change, it is crucial that school curricula stay current with the latest policy initiatives. Schools should adopt and integrate new climate education strategies, ensuring that the material remains relevant and impactful.

3. Enhance local public participation and improve accessibility of relevant information
We suggest these could be done in Hong Kong:

I. Community climate and environment protection societies and public incentives
Aiming at fostering environmental awareness and action, this initiative will involve a high-intensity promotional campaign utilising various platforms such as videos, social media, and posters to reach a broad audience. Registration for participation will be free, with incentives provided to encourage community involvement. A local example of such an initiative is the "GREEN@COMMUNITY," [1]which demonstrates successful community engagement in environmental activities.

II. Promote public awareness raising activities
To motivate residents to partake in environmentally-friendly activities, allowing them to earn credits by participating in events such as upcycling reusable bags is also a crucial action.Workshops will also be organised to teach practical skills related to sustainability, such as upcycling daily waste and creating enzyme cleaners from food waste. Key stakeholders in this initiative will include community managers, the Environmental Protection Department (EPD), and committees from local community centers, all working collaboratively to promote a greener future.

[1] https://www.wastereduction.gov.hk/en-hk/waste-reduction-programme/greencommunity

Climate justice and human rights

Introduction
The intersection of climate change with the rights to health, housing, and information presents pressing challenges in Hong Kong. From escalating health risks due to rising temperatures to inadequate housing conditions exacerbating vulnerabilities, coupled with disparities in accessing climate-related information, the need for comprehensive and inclusive solutions is paramount.

#Right to Health
Climate change is exacerbating the frequency and severity of natural disasters like hurricanes, heatwaves, floods, and wildfires. These events are linked to health impacts such as injury, disability, crop loss, and damage to water or sanitation infrastructure, leading to increased vector-borne diseases. According to a local study by the Chinese University of Hong Kong, when the daily mean temperature exceeds 28.2°C in Hong Kong, the mortality rate increases by 1.8% for every 1°C rise in temperature. By the end of this century, heat-related mortality in Hong Kong is projected to increase by 24% to 2.3 times compared to 2022. The resulting economic losses are also estimated to increase by HKD 11.1 to 110.8 billion compared to 2022.

Protecting human health and well-being from climate change requires universal access to quality essential services without discrimination. According to a 2018 local study by CCOUC, there is room to improve public awareness and action on climate change and related natural disasters and epidemics. Climate plans should be transparently financed and designed through inclusive consultation with affected groups. Impact assessments must ensure climate actions respect all human rights, especially the right to health. A rights-based approach is crucial to strengthen policies at all levels and promote long-term health and sustainable development.

Thus, we demand the authorities to:

1. Reducing residents' exposure by increasing shaded areas on streets and strengthening cooling center services.
2. Bolstering public health policies to better address the mortality risks associated with high temperatures.
3. Develop resilient health systems and infrastructure, including for water/sanitation.
4. Promote universal health coverage and social protection, attentive to vulnerable groups, with accessible early-warning information.
5. Target climate action opportunities that achieve co-benefits, reducing ill-health, building community resilience, and alleviating poverty.

This integrated approach aims to protect human health and well-being in the face of climate change through inclusive, rights-based, and health-focused policies and interventions.

#Right to Housing
In Hong Kong, households living in inadequate housing account for 127,500 units in 2023 (Housing Bureau, 2023). over 220,000 lived in cage homes (Wong, 2022). These impoverished residents are mostly living in subdivided flats, rooftop structures and cage homes, and suffer the most.
Residents are often unable to afford air conditioning, and many rooms are windowless and unventilated. As a result, temperatures are often much hotter than it is outdoors, with such a

difference reaching six degrees Celsius (Lau, 2012). For some of the rooftop structures, residents usually used the metal sheets as their roof and ceiling. This material absorbs and traps heat, causing high indoor temperature at night (Fung, 2024).

The surrounding environment also affects the indoor temperature of the housing. The housing is situated in the city's most densely built-up areas, which are most suffered from the "heat island effect". The surrounding concrete buildings have high heat capacity, absorbing and storing heat. The high-density buildings also block the view of the sky and thus reduce the heat release back to space. The dense buildings also block the air flow and reduce the wind speeds (HKO), which further affect the indoor ventilation.

Improper plumbing is also common, leading to unhygienic conditions. After storms, many residents struggle to afford repairs to their homes as well.

All three of these problems will be exacerbated under climate change as temperatures increase, mosquitos become more active, and storms become more frequent. This does not align with the definition of "adequate housing" as outlined in the United Nations Universal Declaration of Human Rights. It violates the principle of habitability, which is listed as one of the minimum requirements and defined as "[…] protection against the cold, damp, heat, rain, wind, other threats to health and structural hazards (United Nations, 1948)."

Therefore, we demand:

1. The Social Welfare Department extends and expands the "Assistance Programme to Improve the Living Environment of Low-Income Subdivided Unit Households".
    a. The programme ended in June 2022, and we demand it be extended.
    b. The programme provided funds for qualifying families to purchase furniture and undergo simple home repairs. We demand the ceiling of the subsidy be increased. In accordance with recommendations from the report titled "'Impacts of extreme weather on subdivided flat residents" by Oxfam HK (2021), we demand raised floor storage spaces and heat insulating mats to be added to the programme's "prescribed checklist to improvement items" (「改善項目清單」). In addition, we demand the inclusion of Energy Label guidelines in the "prescribed checklist to improvement items".

2. The Housing Authority or Building Departments to implement a registration system for subdivided flats. In accordance with a report by the Hong Kong Institute of Surveyors, owners shall entrust professionals to inspect and accredit the units (RTHK, 2024). Such inspections are to be conducted every three years. Owners of sub-standard units are to be given 12 months to make improvements.

3. The Housing Authority to review the Public rental housing (PRH) allocation mechanism, to consider the priority of the applicants from subdivided flats. As at end-March 2024, the average waiting time for general applicants who were housed to PRH is 5.7 years (Housing Authority, 2024). It is suggested that the applicants from subdivided flats should have the average waiting time shortened to 3 years.

4. The Housing Authority to establish The Adaptation Sub-Committee to perform Climate Change Risk Assessment to assess the vulnerability, adaptability and resilience of planned

area, buildings and the communities to climate change, including but not limited to Risk to health and well-being of household from overheating and extremities in weather, Risk to communities and buildings from flooding, and Risk to coastal communities from sea levels rising.

5. The Housing Authority considers climate resilience in future interim housing, public housing, and light public housing projects, and redevelopment projects. According to the "Practical Guide to Climate-resilient Buildings & Communities" by the UNEP, the following shall be included in designs for hot-humid climates:
    a. Use materials which reduce heat gain, provide fast heat loss and resist moisture,
    b. Ensure openings to buildings avoid direct sun,
    c. Provide green space, access to external shaded space, and water features,
    d. Plan for higher placement of electrical, ventilation and heating systems,
    e. Adopt water-cooled air-conditioning system in public housing estate as suggested by EMSD, and
    f. Use sloped roofs instead of flat roofs (Gupta et al., 2021).

#Right to Information

Make Official Climate Change-related Information More Inclusive

Climate change-related information in Hong Kong lacks inclusivity and accessibility. Despite comprising approximately 8.4% of the population (Home Affairs Department Race Relations Unit, 2021), non-Chinese people face difficulties in accessing crucial climate-related information. Official information about extreme weather events like heat waves, rainstorms, and flooding, as well as information about relevant facilities like temporary heat shelters, are only available in Chinese and English. The language barrier makes ethnic minorities like Indian, Pakistani, Nepalese, Thai, Indonesian, and Filipino, who may not understand Chinese and English, fail to access necessary information to deal with extreme weather and the effects of climate change.

Thus, we demand the government to:-

1. Extreme weather information and relevant health information available in different languages like Bahasa Indonesia, Hindi, Nepali, Punjabi, Tagalog, Thai, Urdu, and Vietnamese.
2. Information about temporary shelters for extreme weather available in different languages
3. At least one staff member stationed at a temporary shelter for extreme weather can offer necessary help for ethnic minorities in their language

Conclusion

Addressing the multifaceted impacts of climate change on health, housing, and information access demands a holistic approach rooted in human rights principles. By advocating for improved living conditions, equitable access to climate information in diverse languages, and robust adaptation strategies, Hong Kong can cultivate a more resilient and inclusive society, ensuring that all residents are empowered to navigate the complexities of a changing climate with dignity and security.

Climate Finance

Introduction

The challenge of mobilizing adequate climate finance has been a central and highly contentious issue in international climate negotiations. At the heart of the debate is the longstanding divide between the global North and global South over the responsibility and burden-sharing for providing the financial resources needed to support climate action in the developing world.

Adding to the complexity, past efforts to leverage private sector investment and innovative financing mechanisms, such as carbon markets, have largely fallen short, underscoring the need for more tangible public policy mandates and financing solutions. The recent announcement of the $1 billion Climate Finance Action Fund (CFAF), to be launched at COP29, represents a potentially important new initiative. Harnessing both public and private capital, the CFAF aims to direct much-needed climate finance towards projects in the developing world.

Recommendations

Currently, Hong Kong focuses on a voluntary carbon market through HKEx's Core Climate platform. However, the city has not yet implemented a carbon pricing mechanism or established rigorous regulations for market participants. Hong Kong's carbon market lacks the transparency and regulatory rigour found in other leading markets, such as those in the EU. The over-the-counter nature of the market makes it less effective in connecting with international standards, and the absence of a carbon price weakens the market's impact on reducing emissions.

1. <u>Carbon tax</u>
   The recommendation is to implement a carbon tax in Hong Kong, allowing companies to offset up to 5% of their taxable emissions using high-quality international carbon credits and China's national carbon credits, following the model used in Singapore. An initial carbon tax rate should be introduced for listed companies, with a gradual increase over time to drive domestic emissions reductions and maintain competitiveness with regions like Korea, Singapore, and California. Exemptions or tax rebates should be offered to SMEs and industries with high emissions but limited ability to pass on costs, while encouraging high-carbon-intensity sectors, such as real estate and logistics, to pilot the carbon tax.

2. <u>Funding to adaptation</u>
   We recommend that the government increase the percentage of proceeds from the Government Green Bond Programme that are earmarked for climate adaptation efforts. The goal should be to strengthen progress towards the Global Goal on Adaptation. According to the latest Green Bond Report 2023, 4% of projects have been channelled towards climate change adaptation. Meanwhile, other countries have recognized the pressing need and have started investing more in adaptation funding. For example, Singapore has recently established a Coastal and Flood Protection Fund with an initial SGD 5 billion allocation. 2023's aftermath of Typhoon Saola and the occurrence of an extremely rare '500-year black storm' in Hong Kong have highlighted the city's lack of preparedness in the face of acute extreme weather events. Given this, it is imperative for the Hong Kong government to increase the financial resources directed towards strengthening the resilience and reliability of the urban infrastructure.

Suggested modification: At COP 29, we urge all parties to ensure climate funds are allocated equitably between mitigation and adaptation for a resilient future. As youth from the global south, we emphasize the need for balanced strategies. For example, Hong Kong's $240 billion climate budget lacks dedicated adaptation funding. We support a review of these strategies to enhance resilience, suggesting that adaptation projects be mandated when issuing corporate green bonds or sustainability-linked loans.

We recommend increasing proceeds from Hong Kong's Government Green Bond Programme for climate adaptation. Currently, only 4% is allocated to adaptation, which is insufficient. Other nations, like Singapore, have recognized this need and established significant funds, such as the SGD 5 billion Coastal and Flood Protection Fund. Recent extreme weather events in Hong Kong highlight the urgent need for more robust urban infrastructure. It's crucial for the government to boost financial resources to enhance resilience.

Conclusion

In all, investing in climate adaptation is crucial for Hong Kong as the city faces increasing threats from extreme weather. Funding should prioritise adaptation, allow developing countries to safeguard its people, economy, and environment in the face of a changing climate.

Implementing an effective carbon tax is another key lever for Hong Kong to drive down domestic greenhouse gas emissions and work toward the Hong Kong Climate Action Plan 20250. A well-designed and transparent carbon pricing system is still yet to be announced in Hong Kong carbon taxonomy and which could be used to incentivize businesses and consumers to reduce their carbon footprints, while generating revenue and create a new stream of business models and which can reinvested into clean energy, emissions reduction programs, and other climate solutions. Coupled with sector-specific emissions targets, a carbon tax can help to steer Hong Kong' s economy and society towards a balance between carbon pricing, emissions goals, and climate adaptation will be crucial for Global south.

Phase out of Fossil Fuels and Renewable Energy Development

Introduction

Following the Paris Agreement signed in 2015, the Hong Kong government established its first Climate Action Plan in 2017, with the latest version published in 2021, stating its climate targets in reaching carbon neutrality by 2050. While the development of renewable energy (RE) has provided opportunities for the transition towards a low-carbon energy system, the energy sector has remained a dominant carbon emitter for the city. At present, less than 1% of total energy produced is made from renewable sources whilst coal and natural gas together account for 75% of the energy mix. Challenges faced by the community include limited space for solar and wind installations in its dense urban environment. Resource constraints and high initial costs for infrastructure add financial hurdles, while complex regulations make adoption difficult. Increasing public awareness and support for renewable initiatives is yet to be accomplished. Addressing these challenges requires innovative solutions and strong policy support to advance towards a sustainable energy future. Therefore, we urge policy makers to consider the following recommendations for implementing RE in Hong Kong.

Recommendations

1. <u>Setting ambitious renewable energy targets and robust roadmap</u>

   Aim for a more ambitious target of at least 20% renewable energy usage by 2030: Currently, the Hong Kong Climate Action Plan 2050 targets 7.5-10% by 2035. Building upon the COP28 pledge to triple renewable capacity by 2030, we urge the Hong Kong government to commit to a minimum of 20% renewable energy usage by 2030. A 2021 study by Hong Kong Baptist University suggests that solar power could meet 21% of our electricity demand by 2030, with other local renewables contributing 14%. Additionally, purchasing renewable energy certificates and importing renewable energy from the Greater Bay Area could cover about 17.7% of our demand. (Mah et al., 2021; WWF Hong Kong, 2017).

   Establish a robust roadmap: This includes setting milestones for the short term (2025-2030), medium term (2030-2035) and long term (2035-2050), with specific targets and action plans such as pilot solar power and offshore wind projects, infrastructure upgrades, and policy refinement.

2. <u>Engagement with the private sector to accelerate the deployment of renewable energy</u>

   According to CLP Climate Vision 2050, 55% of generation and energy storage capacity from nuclear and renewable sources is projected in 2030 (CLP Power, 2024). We urge CLP to increase the proportion of electricity generation from renewable energy, including solar, wind, and waste-to-energy.

   On the other hand, the Water Supplies Department (WSD) has implemented pilot schemes for floating photovoltaic (FPV) systems at several reservoirs since 2017 (WSD, 2024). It is

recommended that the private sector provide technical or financial support to expand the schemes to a larger scale by 2030.

BEAM Plus should also become a mandatory measure to develop local buildings sustainably, such as offices and shopping malls. Currently, the Hong Kong Green Building Council launched it only as a voluntary rating scheme (Hong Kong Green Building Council Limited, 2024). The government should review BEAM Plus and the building energy code and launch a mandatory scheme.

3. Maximise the use of existing resource and infrastructure
   At present, O · PARK1 is capable of handling a maximum of 200 tonnes of food waste daily for power generation (Opark1, 2024). Meanwhile, due to high logistics costs and difficulties in collecting domestic and commercial waste this facility is not operating at its full capacity. It is essential to focus on improving efficiency of available infrastructure through introducing official schemes to encourage public and business participation, such as giving incentives and setting mandatory standards to follow. On the other hand, despite current efforts in developing hydrogen as an alternative for natural gas, technical issues such as storage of low-density gas and safety concerns are some challenges to overcome as a globe. Therefore, it seems to be more practical to reallocate resources to more mature and proven technologies, owing to the fact that Hong Kong is currently falling behind its schedule in the Climate Action Plan 2050 (EPD, 2021).

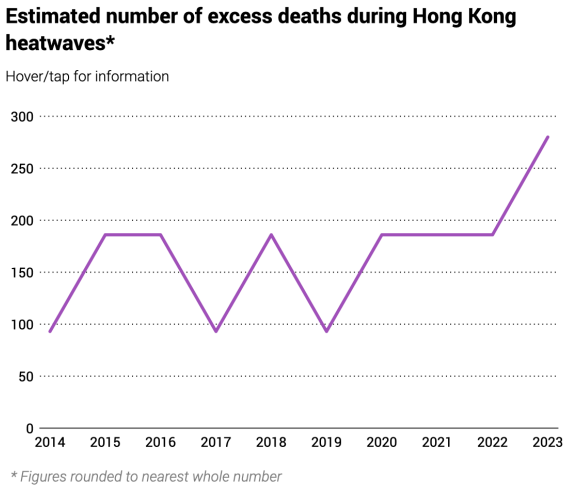
Conclusion

Addressing climate change requires more than merely establishing targets for renewable energy's share in the energy grid by a specific year. A successful strategy should incorporate a solid roadmap, foster public-private partnerships, and effective resource allocation. By implementing these measures, we are confident that we can progress toward achieving a net-zero society by 2050, with renewable energy playing a predominant role in our energy grid.

Climate Resilience

Introduction

Problem statement: Underestimated needs of social-economically disadvantaged food delivery workers under extreme climate and weather conditions



Researchers from the University of Hong Kong found that each heatwave, defined as periods when daily temperatures exceed 30.6°C (87.1°F), resulted in about 93 additional deaths. They analyzed excess deaths over a 20-day period, ensuring a minimum 20-day gap between the start and end of their study period.

Preambulatory Clauses:

1. *Recognizing* that food delivery workers often work long hours in challenging conditions, including extreme heat, cold, and inclement weather;
2. *Acknowledging* that the physical demands and environmental stresses experienced by food delivery workers can have significant impacts on their health and wellbeing;
3. *Concerned* that the needs of socially and economically disadvantaged food delivery workers may be underestimated or overlooked by employers and policymakers;

Recommendations:

1. Policy changes:
    a. Acknowledge the dangers of outdoor work
    b. Simplify the process of bringing claims to Labor Tribunal
    c. Relaxed the delivery time requirement from 30 minutes to 1 hour (allow buffer time for delivery).

2. Strengthen labor protections including guaranteed rest breaks, access to protective equipment.
    a. Legislate food delivery provider compensating accident, third party liability and heatwave insurance for self-employed courier under extreme heat weather

3. Establish targeted social welfare programs and income supports for low-wage workers in the food delivery sector.
    a. Advocate and lobby for a legislative agenda for providing extreme weather contingent and remedy
    b. Provide unemployment insurance

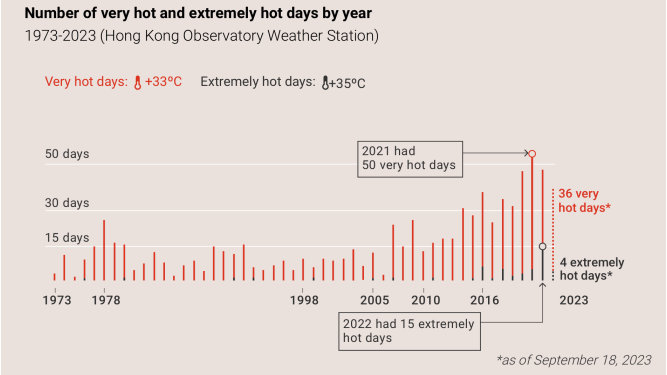
4. Provide a uniform with anti-heat fabric(Protective Clothing)

a. Data transparency: platform connects with Hong Kong Observatory to adapt cooler delivery routes.
   =>Show hot districts
b. Improve the deliveryman's uniform material and equipments
c. Equip workers with lightweight, breathable, and UV-protective clothing to minimize heat absorption and prevent sunburn.

5. Create an advisory board to collaborate with community organizations to understand the unique challenges faced by marginalized food delivery workers.
   a. Invest in climate adaptation to improve working conditions, such as providing water stations, and shelter resources during heat waves.
   b. Heatwave Alerts:
      i. Implement a system to distribute heatwave alerts and safety tips through mobile apps or SMS, keeping delivery personnel informed and prepared for extreme weather.

6. Health Monitoring Wearables:
   a. Deploy wearable technology that monitors vital signs, such as heart rate and body temperature, providing alerts to both workers and managers about potential health risks.

Research-Based References



Examples from overseas:
USA: Shutdown (app) operation system at impacted areas due to extreme weather conditions

Conclusion & Summary of Recommendations

1. Acknowledged outdoor work dangers, simplified tribunals claims process, relaxed 30-min delivery requirement to 1-hour.
2. Strengthened labor protections, introduced insurance for couriers in extreme heat.
3. Provided social welfare, unemployment insurance, cooling uniforms, data-driven route optimization.
4. Form advisory board, invested in climate adaptation like water stations and shelters.
5. Shutdown or Limit operations when Labour Department issues the Black Heat Stress at Work Warning
6. Provide government subsidies to delivery drivers on an ad hoc basis.

Technology and Innovation

Introduction

Hong Kong's geographical positioning leaves it extremely vulnerable to the effects of climate change. Therefore it stands to gain a lot from leveraging technology and innovation for climate mitigation efforts and to increase its climate resilience, such as through utilising geospatial data for better urban planning.

However, a brain-drain of talent as well as barriers to commercialising innovative research into technologies applicable to the real-world are key problems that Hong Kong must address to effectively leverage technology and innovation to address the climate crisis.

Recommendations

1. <u>Develop a robust talent pipeline</u>
   A strong technology and innovation ecosystem is underpinned by talents who have the knowledge, skills and capabilities to develop new technologies that address pressing problems such as climate change. In particular, it is recommended that Hong Kong particularly seek to develop Science, Technology, Engineering, Arts and Mathematics [STEAM] skills in both its secondary and tertiary education students. As noted by Big Bang Academy (2024), developing STEAM skills plays an important role in fostering scientific, technological, social and economical innovation. Indeed, Singapore's strong technology and innovation ecosystem has often been credited to its well-developed tech talent in its workforce.

2. <u>Increase awareness and training for researchers to commercialise innovative research</u>
   In order to address the gap between research and its translation into scalable market-ready technologies, we recommend that Hong Kong invests resources into raising the awareness of commercialisation pathways for researchers while also strengthening the regulatory framework around intellectual property to address concerns some researchers may have when considering commercialising their technologies. In addition, rather than relying on just knowledge transfer offices to bring innovations from universities to market, we recommend that more education and training be provided directly to researchers so they have the skills and confidence to consider building sustainable startups from research themselves.

3. <u>Utilise relevant technologies to increase climate resilience</u>
   Hong Kong's urban planning struggles demonstrate the need for more comprehensive, long-term strategies that leverage geospatial data. The city's overreliance on land reclamation, which faces significant environmental and logistical hurdles, contrasts with greener alternatives like utilizing brownfield sites and agricultural land. Geospatial data integration is a core strength of Geographic Information Systems [GIS], enabling urban planners to combine diverse data sources into a comprehensive location-based system. This empowers complex spatial analysis and a deeper understanding of urban dynamics, from mobility patterns to community development needs.
   Beyond urban planning, the document underscores the key role of GIS in climate change modelling and analysis. Geospatial technologies arm policymakers and the public with vital insights, from mapping greenhouse gas emissions to predicting sea level rise impacts on

coastal communities.

Geographical Information Systems (GIS) play a pivotal role in climate change modeling and analysis. GIS is a tool that allows us to visualize, analyze, and interpret data to understand patterns, trends, and relationships related to location. When it comes to climate change, GIS is used to model and analyze diverse and complex climatic data which can be used to develop robust disaster management plans and mitigate crises. Such clear visualizations are game-changing regarding accurately identifying environmental hazards and at-risk areas. For example, NASA used GIS in its NASA Disasters Mapping Portal during wildfires across in 2020 to help first responders handle simultaneous wildfires and assist disaster coordinators in pinpointing active fire points, fire modeling, and analyzing smoke plumes and air quality. By visualizing climate data and environmental challenges, GIS facilitates informed decision-making and collaborative efforts to build a more climate-resilient Hong Kong.

Conclusion

Technology and innovation will play a key role for Hong Kong as the city faces increasing threats from extreme weather. In order to strengthen the foundations for a dynamic technology and innovation ecosystem, a greater emphasis on teaching STEAM skills in secondary and tertiary education is recommended to develop a strong talent pipeline and there should be increased guidance and training for researchers on how to commercialise innovative technologies originating from research. Innovative technologies such as Geographical Information Systems (GIS) should be used for climate change modeling and analysis to improve Hong Kong's climate resilience in the face of the climate crisis.

Nature, Biodiversity and Oceans

Introduction

Despite being a 'concrete jungle', Hong Kong has a wide range of habitats which nurture rich biodiversity. For instance, the area of country parks and special areas occupies about 40% of total territory area (Agriculture, Fisheries and Conservation Department, 2020). However, Hong Kong is facing a threat of habitat loss, mainly due to the insufficient protection offered by the existing regulations, and the direct destruction caused by the rapid urban development.

The implementation of the Northern Metropolis Development Strategy (HKSAR, 2021) further exacerbates the pressure on agricultural land and natural habitats. This strategy, while aimed at urban development, poses significant risks to the remaining natural areas and the biodiversity they support.

To address these challenges, it is crucial to enhance conservation efforts and integrate sustainable practices into urban planning. By doing so, we can ensure that Hong Kong's unique biodiversity is preserved for future generations.

Recommendations

1. <u>Align with International initiatives and policies</u>
   The Convention on Biological Diversity (CBD) has set an ambitious target of protecting 30% of land and ocean by 2030. This global initiative underscores the urgent need for comprehensive conservation efforts. While Hong Kong's country parks contribute significantly to land conservation, covering about 40% of the territory, marine protected areas and reserves account for less than 5% of Hong Kong waters. To align with CBD targets, it is essential to expand marine and coastal protected areas to cover up to 30% of Hong Kong waters. This expansion along with country parks should be coupled with effective management strategies to ensure the sustainable use and protection of these natural resources. Proper management includes enforcing regulations, preventing illegal activities, and promoting sustainable practices among local communities and industries.

2. <u>Land use and conservation</u>
   Regular monitoring and patrolling are crucial to ensure the effectiveness of conservation efforts. This includes regular assessments of land use types and the health of ecosystems. Such monitoring helps identify changes or threats to biodiversity and informs timely policy interventions. This ensures that conservation strategies are based on accurate and up-to-date information, allowing for adaptive management approaches that can respond to emerging challenges. Engaging local communities in monitoring and conservation efforts can enhance the effectiveness of these initiatives. Community-based monitoring programs can provide valuable insights and foster a sense of stewardship among residents.

3. <u>Incorporation of Nature-based Solutions (NbS)</u>
   Incorporating NbS into climate strategies is essential for both mitigation and adaptation. Case studies from other regions, such as the EU Nature Restoration Law, can serve as valuable references for setting restoration goals and integrating NbS into urban planning and design. For example, restoring wetlands, oyster reefs, and coral communities can significantly

enhance coastal defense by attenuating wave energy. These natural barriers reduce the impact of storm surges and coastal erosion, protecting coastal communities. Additionally, wetlands and urban parks could play a crucial role in carbon sequestration, capturing and storing carbon dioxide from the atmosphere. This not only helps mitigate climate change but also enhances biodiversity and ecosystem resilience.

4. Use of natural area
   Native plants attract and support pollinators such as bees, butterflies, and birds. These pollinators are essential for the reproduction of many plant species and the overall health of ecosystems. Green spaces, including parks, rooftop gardens, and green corridors, offer suitable habitats for various urban wildlife. These areas can support species such as birds, insects, and small mammals, promoting biodiversity even in densely populated cities. Green spaces with native vegetation can help mitigate urban heat island effect by providing shade and through the process of evapotranspiration, where plants release water vapor into the air, cooling the environment. This can significantly lower urban temperatures, making cities more comfortable and livable.

5. Publicise environmental education
   In the current school curriculum, nature and climate education are very limited, which is a significant oversight given the importance of these topics. Acquiring knowledge is the first step toward effective conservation actions, as it equips students with the understanding and skills needed to address environmental challenges. Therefore, we recommend that the Education Bureau enhance student climate and environmental literacy by integrating comprehensive environmental education into all levels of schooling. This should include providing training courses for teachers to impart relevant knowledge to young minds, and proposing more compulsory yet practical events for pre-teens, such as climate lectures, field trips to organic farms, and beach-cleaning events. Additionally, offering workshops for the elderly, including climate education dramas and volunteering opportunities with NGOs, can foster a community-wide commitment to sustainability and conservation.

Conclusion

Hong Kong's rich biodiversity is under significant threat due to habitat loss and rapid urban development. To address these challenges, it is crucial to align with international conservation targets, such as the CBD's goal of protecting 30% of land and ocean by 2030. Expanding marine and coastal protected areas, coupled with effective management strategies, is essential. Regular monitoring and community engagement are vital for the success of conservation efforts. Incorporating Nature-based Solutions (NbS) into climate strategies and utilizing native species in urban greening can enhance biodiversity, climate resilience, and the overall quality of urban environments.

Politics, Policymaking, and Governance

Introduction

1. Low public participation in policy making
   Hong Kong faces low engagement from sustainability-interested individuals in policy making, stemming from limitations in the local public consultation process:

   I. Research on rural tourism development revealed 68.7% of respondents (n = 241) felt uninvolved in decision-making. Top reasons included availability, age, unclear roles, and criticism of previous consultations (Mak et al., 2017).
   II. The government "streamlined" land development consultation process reduces public participation opportunities, reinforcing perceptions of tokenism and eroding public trust in the impact of their input.
   III. For the last policy address (2023), the government conducted a one-off, formal public consultation session targeting post-secondary school students (The Government of HKSAR, 2023).

2. Specialisation of Government departments
   The Hong Kong Government has reformed its civil service system with New Public Management (NPM), an approach that emphasizes specialization and efficiency (Berman, 2010). This reform has strengthened cooperation between government and private sectors, leading to the outsourcing of many public services. Consequently, civil service work has become more specialized with less interdepartmental cooperation (Berman, 2010).

   Notably, Hong Kong's Climate Action Plan 2050 provides limited discussion on departmental responsibilities in addressing climate change (Hong Kong Government, 2021). The combination of unclear responsibilities and NPM reforms may lead to a 'diffusion of responsibility' phenomenon (Bickman, 1972), hindering environmental goal achievement.

Recommendations

1. Comprehensive Consultation
   We urge the government to employ flexible, innovative public consultation methods to increase stakeholder engagement in sustainability and climate change policy-making. This approach would also demonstrate the government's willingness to consider public input.

   Successful examples:
   Overseas: ChangeExplorer (UK) - A smartphone app using location data to notify users about local redevelopment plans (Wilson et al., 2019).
   Local: PRAISE-HK - Empowering public participation in air quality and health risk discussions.

   Recommended innovative formats include:

   - **Climate Resilience Design Thinking Workshops** and **Hackathons**: Bringing citizens together to develop innovative climate policy solutions.

- **Virtual Reality (VR) Simulations** and **Augmented Reality (AR)**Climate Tours: Helping people experience potential climate scenarios.
- **Gamified Apps**: Educating and involving the public in climate action.
- **Cross-generational Climate Dialogues** and **Role-Play Simulations**: Facilitating understanding among diverse stakeholders.

These methods aim to increase engagement, gather diverse perspectives, and foster collaborative solutions. The most effective ideas should be incorporated into policy-making, ensuring climate strategies are well-informed, widely supported, and adaptable.

2. Interdepartmental Coordination Mechanism
We propose developing an **interdepartmental coordination mechanism** within the Hong Kong Government (Shulaev, Sabitova, & Berheeva, 2022) by:

- **Increasing the number of accountability officers** in the Council for Carbon Neutrality and Sustainable Development to address the 'Diffusion of responsibility' phenomenon (Bickman, 1972).
- Clearly measuring responsibilities using the "SMART principle" to increase self-efficacy among committee members and benefit the achievement of Climate Action Plan 2050 goals (Doran, 1981).
- Integrating each committee member's responsibilities and goals into performance evaluations to strengthen individual accountability (Usrey, Palihawadana, Saridakis, & Theotokis, 2020). Members who fail to meet their goals may be required to provide explanations.

These measures aim to enhance interdepartmental coordination and individual accountability in achieving climate-related goals.

Action Items

1. Introduce innovative public consultation formats:
    a. Climate Resilience Design Thinking Workshops and Hackathons
    b. Virtual Reality (VR) Simulations and Augmented Reality (AR) Climate Tours
    c. Gamified Apps for climate education and action
    d. Cross-generational Climate Dialogues and Role-Play Simulations
2. Enhance the Council for Carbon Neutrality and Sustainable Development:
    a. Increase the number of accountability officers
    b. Implement SMART principle for measuring responsibilities
    c. Integrate climate-related goals into performance evaluations
3. Improve interdepartmental coordination:
    a. Clearly define departmental responsibilities for climate action
    b. Establish mechanisms for regular cross-department collaboration on climate initiatives
4. Reform public consultation process:
    a. Diversify consultation methods beyond traditional formats
    b. Ensure wider demographic representation in consultations
    c. Provide clear feedback on how public input influences policy decisions
5. Strengthen individual accountability:
    a. Develop a system for tracking and reporting progress on climate-related goals
    b. Implement regular reviews and adjustments of climate action strategies

6. Foster long-term vision:
    a. Develop a comprehensive plan for Hong Kong to become a global leader in urban sustainability
    b. Integrate climate resilience and sustainable development into all aspects of city planning and governance

Conclusion

The challenges of low public participation and government department specialization present significant obstacles to Hong Kong's sustainable development, but also opportunities for innovative solutions. By implementing comprehensive consultation methods and establishing an interdepartmental coordination mechanism, Hong Kong can foster a more engaged citizen participation and a responsive government. These actions represent a fundamental shift towards participatory and integrated governance.

The long-term vision for Hong Kong should be one of a city that leads by example in climate resilience and sustainable development. This encompasses an engaged populace, collaborative government structure, and a thriving green economy. By taking decisive action now, Hong Kong can transform these challenges into opportunities, paving the way for a more sustainable, resilient, and prosperous future as a global leader in urban sustainability.

Circular Economy

Introduction

The traditional linear economic model in Hong Kong, which is based on a take-make-consume-throw away pattern, creates substantial waste. The Hong Kong government aims to promote the gradual development of a circular economy and reduce reliance on landfilling (Environment Bureau, 2021, p.4). The current waste reduction action plan focuses much on recycling and waste-to-resources/energy (Environment Bureau, 2021, pp.17-18). Mandatory producer responsibility schemes are based on the polluters-pay principle and focus on requiring stakeholders to share the responsibility of dealing with end-of-life products (Environmental Protection Department, 2021). The few strategies to reduce waste generation at the outset are means of education and publicity (Environment Bureau, 2021, pp.7, 20, 23) with no binding effect.

The global market prices of many recyclables decline, and the Mainland as well as other parties to the Basel Convention have tightened up the import control of recyclables (Environment Bureau, 2021, p.4). Therefore, Hong Kong needs a comprehensive waste reduction strategy to ensure that its recycling and waste disposal capacity is not overloaded.

Avoiding and minimizing the generation of waste is central to any waste reduction strategy (Environmental Protection Department, 2022). Producers have a role in waste reduction, such as to offer products with simple packaging, produce products with recyclable materials, etc. (Environment Bureau, 2021, p.16) This was identified as a challenge to developing a circular economy (Environment Bureau, 2021, p.16). If Municipal Solid Waste (MSW) charging had been implemented, households' and enterprises' ability to reduce waste would have been limited by options available for consumption and procurement along supply chains.

To develop a circular economy, move away from landfilling and achieve the net zero targets, the government needs a holistic resource management approach. In order to successfully relaunch MSW charging, comprehensive mandatory producer responsibility schemes should be in place to minimize waste generation throughout products' life cycles.

Recommendations

1. <u>Refining the Producer Responsibility Schemes (PRS)</u>
   To support Hong Kong's environmental goals, consider refining the PRS in three key areas: focusing on the design stage, strengthening current PRS frameworks, and adopting Advanced Fee Modulation.

   Focus on the Design Stage: While current PRS frameworks focus on the end-of-life stage of a product's lifecycle, it is suggested that policymakers encourage Design for Environment (DfE) practices. By addressing environmental factors early in the product design process, manufacturers can enhance recyclability and reduce waste.

   Strengthen Current PRS Frameworks: While PRS in Hong Kong cover selected items like plastic bags, beverage containers and electronic waste (*Producer Responsibility Schemes*, 2021), expanding these frameworks could greatly benefit environmental efforts. Research shows that mandatory Fee Modulation systems are more effective than voluntary ones due to better enforcement and industry fairness (OECD, 2024, 8). Expanding mandatory coverage

could enhance environmental protection.

Adopt Advanced Fee Modulation: Considering the transition to Advanced Fee Modulation could provide more precise incentives for DfE (Laubinger, et al., 2021, 9). This system adjusts fees based on factors such as recyclability and durability, encouraging more environmentally friendly product designs. For instance, Belgium's lower fees for transparent PET, compared to colored PET and Estonia's example of exempting reusable packaging (Laubinger, et al., 2021, 19) highlights further potential benefits.

Additionally, exploring product standards and content requirements similar to the EU Single-Use Plastics Directive could further instigate DfE. These complementary measures would help reduce reliance on non-recyclable materials and support sustainable alternatives. Implementing these suggestions could significantly encourage corporate innovation in DfE, and achieve waste reduction at source.

2. <u>MSW charging</u>
   MSW charging is an effective policy tool to drive enterprises and the public to practice waste reduction and recycling (Environmental Protection Department, n.d.). Food waste, plastic and paper are three major MSW sources (Environmental Protection Department, 2023). MSW charging is necessary to reduce all types of waste, especially non-recyclable food waste. PRS with the proposed changes which address waste reduction starting from the design stage would enable households and enterprises to actively reduce waste through choosing better-designed products in terms of recyclability, packaging, etc. This would ease public concern over the fairness of MSW charging.

3. <u>Policy goals and timeline</u>
   - Continually expand mandatory coverage of PRS to more types of items and materials.
   - Relaunch MSW charging by 2027.
   - Conduct policy research on Advanced Fee Modulation. Launch pilot schemes in selected items and materials by 2030.
   - Expand PRS to all product packaging by 2035 to align with the Waste Blueprint of Hong Kong 2035.

Conclusion

The need for waste reduction is becoming ever more imminent as our three existing landfills are expected to be filled up in the coming years (Environmental Protection Department, n.d.). A holistic approach is necessary to prevent overloading local capacity of waste management strategies and to maximize resource efficiency in view of net zero targets. Progressive policymaking to encourage stakeholders' collaboration and innovation can empower and incentivize enterprises and individuals to reduce waste and make proper use of resources. It is anticipated that society can gradually develop models of production and consumption which yield maximal benefit out of existing resources throughout their extended life cycles. Only then can Hong Kong realize the vision of circular economy and zero landfilling.

Acknowledgments

Action for Climate Empowerment	Ma, Tsz Ching; Chow, Wai Ki; Tang, Wan Yu; Tsoi Nga Sze Kerise
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Climate Finance	Tong, Wei Shin (Alexander); Yung, Wai Lun
Phase out of Fossil Fuels and Renewable Energy Development	Chan, Cheng Long; Ho, Cheuk Ming; Ip, Chun Kit; Leung, Ho Ting; Yuen, Chun Lee Lily
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About LCOY HK 2024

CarbonCare InnoLab (CCIL) is an independent non-government organisation dedicated to the nurturing and development of innovative solutions in response to today's climate change and sustainability challenges. We put special emphasis on multi-stakeholder engagement and solution-oriented processes, as well as awareness-building across the community in Hong Kong. There is a need to motivate communities, and especially young people, to view the transition to a zero-carbon economy as an exciting opportunity rather than a disturbing threat. CCIL turns pessimism about climate change into an optimistic vision of a more engaged and participatory society, and a more resilient and sustainable economy in Hong Kong. There is a way to promote a picture of the future which contains an array of possibilities and opportunities. CCIL carries the news that people can be part of the solution to climate change, not part of the problem.

Our mission is to encourage innovation that is both relevant to local needs and which contributes to solving global climate change challenges. This includes climate justice, carbon reduction, resource conservation and action that will assist adaptation and resilience building. CCIL is a charitable body registered in Hong Kong, enjoying tax exemption status under Section 88 of the Inland Revenue Ordinance.

Generation Climate Hong Kong (GenC) is a youth-led organization dedicated to empowering Hong Kong's young generation and establishing glocal connections for future climate leaders. They aspire to facilitate knowledge exchange, collaboration, and engagement with global climate leaders. Through the network, they aim to amplify diverse voices and perspectives, inspiring and equipping young individuals to drive meaningful change both locally and on the international stage.

Hong Kong Youth For Climate Action (HKYCA) is a youth climate action network to empower young people in Hong Kong to advocate and build creative solutions for international and local climate issues. Their vision is to mobilize Hong Kong youth to take climate action in achieving Paris Agreement Goals. HKYCA provides a youth-led dedicated platform for young advocates to discuss and inspire original insights on climate issues that matter most to them.

Network of Environmental Student Societies (NESS) is a youth-led climate-focus organisation based in Hong Kong. Founded in June 2021, NESS aims to provide a one-stop platform for awareness raising, capacity building, connection development and driving policy changes. With a focus on the interdisciplinary nature of climate change and all things environmental, NESS hopes to inspire youth from all kinds of backgrounds to take solid actions.

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